

Practical work 01 – September 15 2026

Let's get started

The main objective of this Practical Work (PW) is to set up your computer environment. We then have some exercises on **Fundamentals on Machine Learning**. Optional exercises are also provided to refresh students on the use of Python.

You have basically 3 ways to work with Python :

- Using a Python console in interactive mode. When Python is installed on your computer, just open a terminal and launch python.
- Using a text editor or an IDE to develop scripts of functions (i.e. saving ".py" files), as you would do it in Java or C. A good IDE is Pycharm from Jet Brains.
- Using iPython notebooks via a web browser or an IDE where you can mix code and Markdown text inputs in cells.

For this class we recommend you to use option c) above and work with iPython Notebooks for the practical works. It will allow you to report on your experiments and input code at the same time.

Notes : We use Python version 3 in this class. From next week, we will assume that your environment is all set up and that you have gotten familiar with Python.

All PWs are optional but you can submit your PW reports on Moodle for the **next Mondays**. We may override this rule, so pay attention to the dates indicated in Moodle.

Exercise 1 Python installation on your computer

Skip this part if you are already set up with Python, Jupyter notebooks and your favorite IDE for Python. Otherwise, we recommend one of the following installations :

- Jupyter notebooks with manual installation in virtual environments (**recommended**) - First install Python 3 (from 3.10 or up, but python3) from <https://www.python.org>. Then create a working directory and open a terminal in the directory :
 - Install the virtual env : `python3.13 -m venv venv`
 - Activate the virtual env on Mac : `source venv/bin/activate`
 - Activate the virtual env on Windows : `C:\> <venv>\Scripts\activate.bat`

- d) Install Jupyter : `pip install jupyter`
- e) Install needed libs : `pip install numpy matplotlib pandas scikit-learn`
- f) Launch Jupyter : `jupyter notebook`

Whenever you need to re-launch the notebooks, repeat steps b) and f). Whenever needed install requested libraries with `pip install LIB_NAME` once the virtual env is setup step b).

- A recent alternative to `pip` is `uv`. See the doc under <https://docs.astral.sh/uv/>. A bit similar procedure as `venv` and `pip` as above but faster.
- Anaconda platform - a distribution of Python with popular data science packages that are pre-installed or easily installable. <https://www.anaconda.com>. Select the Anaconda Distribution. You should obtain something similar to Figure 1 below (Fig may be outdated).

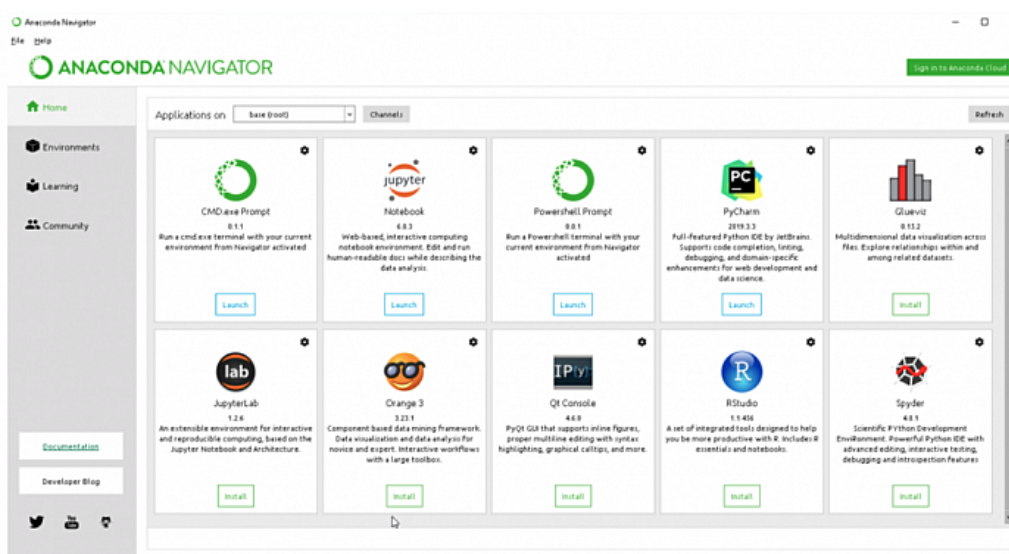


FIGURE 1 – Installing the Anaconda platform

- PyCharm IDE - an integrated development environment for Python. <https://www.jetbrains.com/pycharm>. You may select Professional or Community edition - the Community edition should be enough for this class. Free licenses are given to students, use your `hes-so.ch` address for the registration by JetBrains.
- Visual Studio Code - an IDE from Microsoft that support many languages. <https://code.visualstudio.com/>.
- If you do not want to install anything on your computer, you may use Google Colab at <https://colab.research.google.com/>.

Exercise 2 Python language in a nutshell

We assume here that students are knowledgeable in other programming languages such as Java or C and that basic data structure concepts are known. If you know already Python and the concept of notebooks, then you can skip this exercise.

- Open Jupyter from the command line or from the Anaconda Navigator. Jupyter Python notebooks run in your browser so, after launching Jupyter from Anaconda, a browser should show up. Open a new notebook from menu File and follow the User Interface Tour as illustrated in Figure 2.

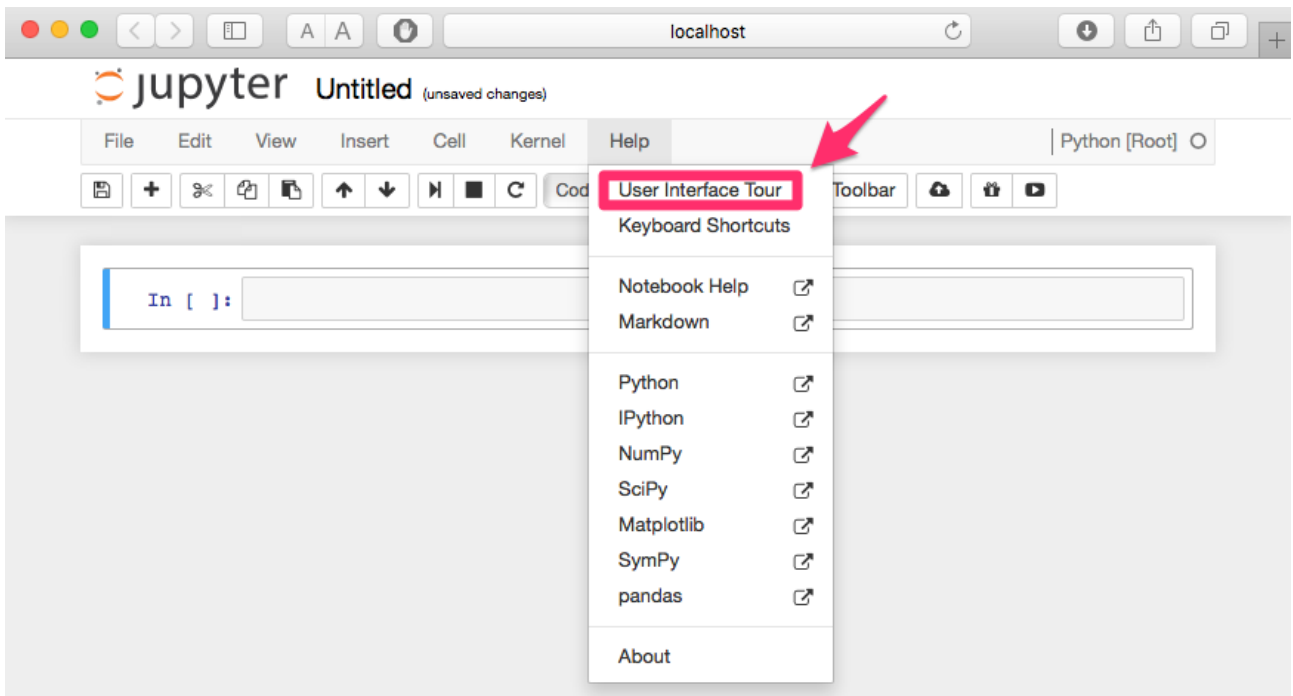


FIGURE 2 – First steps with Python notebooks.

- Download the file `intro-python-3.ipynb` from moodle and open it from Jupyter in Anaconda. You need to navigate where you stored the ipynb file. See Figure 3 below.

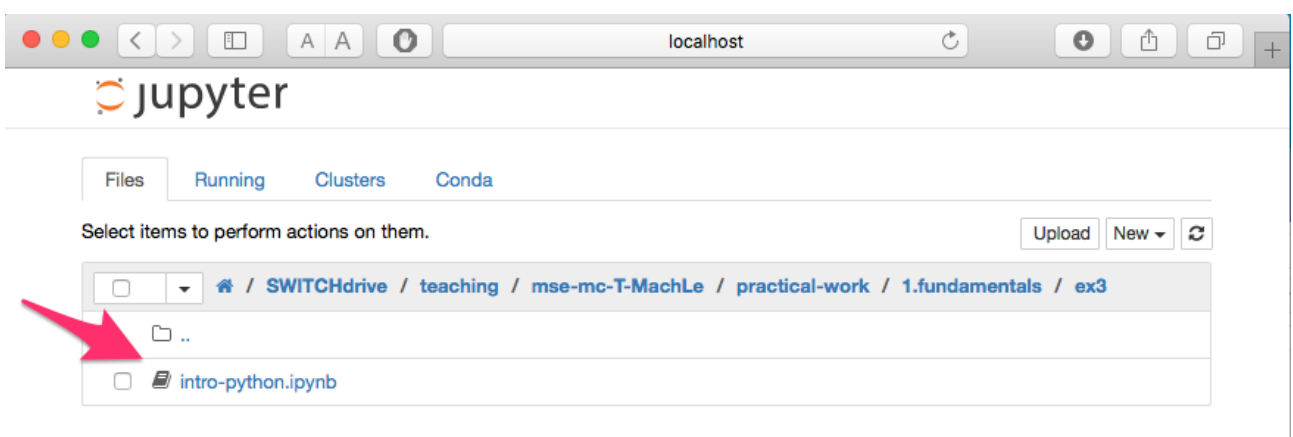


FIGURE 3 – Intro to Python language and Python notebooks.

- Go through the content of the `intro-python-3.ipynb` notebook and play with the cells. This document should give you a quick introduction to Python assuming that you are fluent with other programming languages. For the text cells, you may also want to get

- familiar with Markdown syntax if not known already, for example by reading this page <https://help.github.com/articles/basic-writing-and-formatting-syntax/>.
- If you want a more fully fledged introduction to Python, read the official tutorial from <https://docs.python.org/3.13/tutorial/>.

Exercise 3 Data visualization

To train a bit yourself, we propose you to do a visualisation workout with the Iris dataset https://en.wikipedia.org/wiki/Iris_flower_data_set. Download the file `iris.txt` from moodle or from <http://www.statlab.uni-heidelberg.de/data/iris/>. Put it in a Python data structure and attempt to reproduce a plot close to the one in Figure 4. Advice : start by working with individual plots and then move to a grid of plots with `subplot` or take the shortcut of using the library `pandas`.

The iris species classification task is a classical one. The goal is to distinguish three kinds of iris : setosa, versicolor and virginica. In the data set, a botanist has extracted 4 *features* : sepal length, sepal width, petal length and petal width.

What can you say regarding class separation? One class seems easier to distinguish from the others, which one? How would you separate the other two?

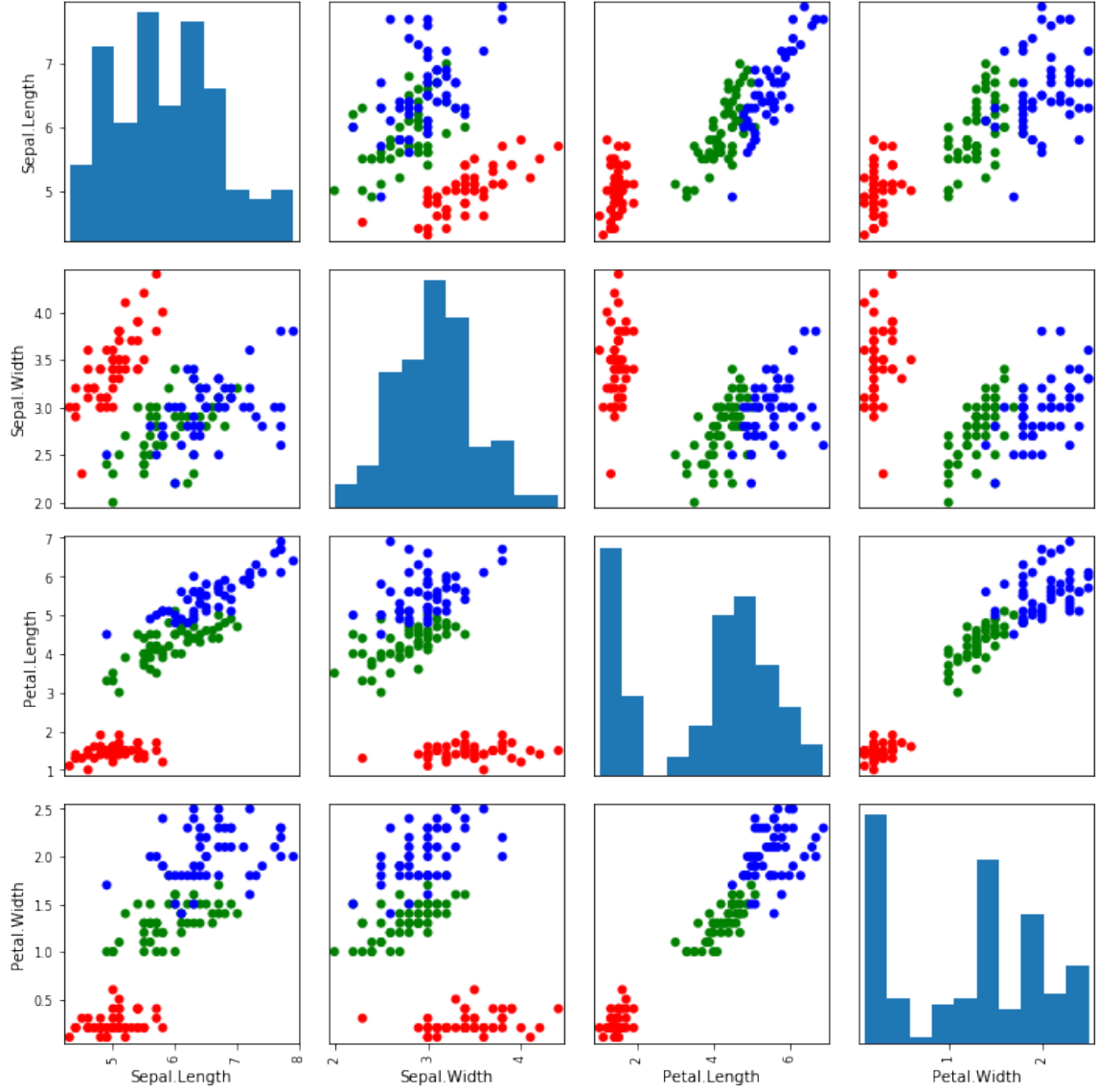


FIGURE 4 – Visualization of the Iris data as a pairwise scatter plot. The diagonal plots the marginal histograms of the 4 features. The other cells contain scatterplots of all possible pairs of features. Red circle = setosa, green diamond = versicolor, blue star = virginica.

Exercice 4 Find your own examples of Machine Learning Tasks

Complete the Table below with 2 new examples of machine learning tasks, explaining what is the task T , performance measure P and training experience E .

Tableau 1-1. Quelques exemples de machine learning proposés par Mitchell

Cas d'application	Checkers learning	Handwriting recognition	Robot driving learning
Tasks T	Playing checkers	Recognizing and classifying handwritten words within images	Driving on public four-lane highways using vision sensors
Performance measure P	Percent of games won against opponents	Percent of words correctly classified	Average distance traveled before an error (as judged by human overseer)
Training experience E	Playing practice games againsts itself	A database of handwritten words with given classifications	A sequence of images and steering commands recorded while observing a human driver

FIGURE 5 – Source : Data Science - fondamentaux et études de cas, Michel Lutz et Eric Bernât, Eyrolles.

Exercice 5 Review questions

1) Supervised vs. unsupervised systems

Of the following examples, which one would you address using a supervised or an unsupervised learning algorithm ? Give some explanations for your answers.

- Given email labeled as spam/not spam, learn a **spam filter**.
- Given a set of news articles found on the web, group them into sets of **related articles**.
- Given a database of customer data, automatically discover **market segments** and group customers into different market segments.
- Given a dataset of patients diagnosed as either having **glaucoma** or not, learn to classify new patients as having glaucoma or not.

2) Classification vs. regression systems

Can we transform a regression problem into a classification problem ? What would be the benefits of doing so ?

3) Other questions

- a) Why is it important that a test set is *independent* to the training set in a machine learning system?
- b) What is the meaning of the hat in the equation $\hat{y} = h(\mathbf{x})$?
- c) What is the difference between machine learning and statistics?
- d) Is a *detection* system a regression or classification system? Give an example.

Exercise 6 Reading assignments

- Read the first chapter of Murphy’s book “Machine Learning”.
- Read the first chapter of Biernat and Lutz’s book “Data Science – fondamentaux et études de cas”.

Build your own summary of these chapters by doing a taxonomy of machine learning problems. You can find the pdfs on Moodle.